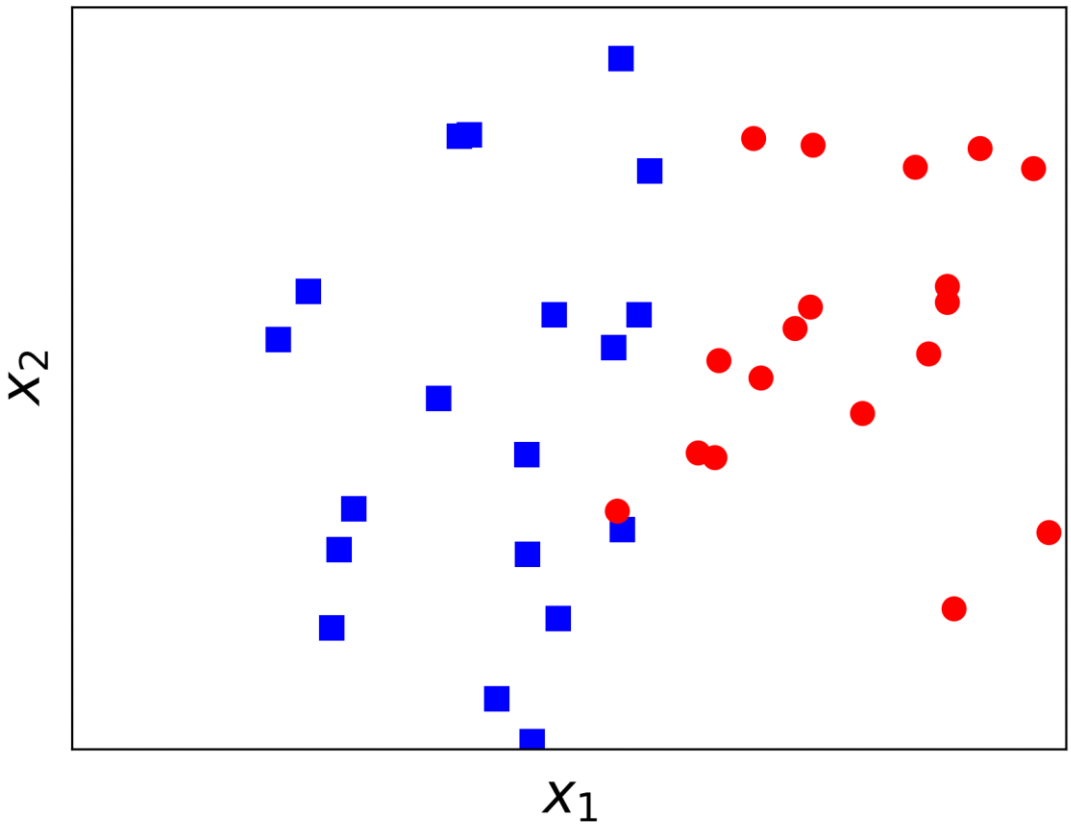
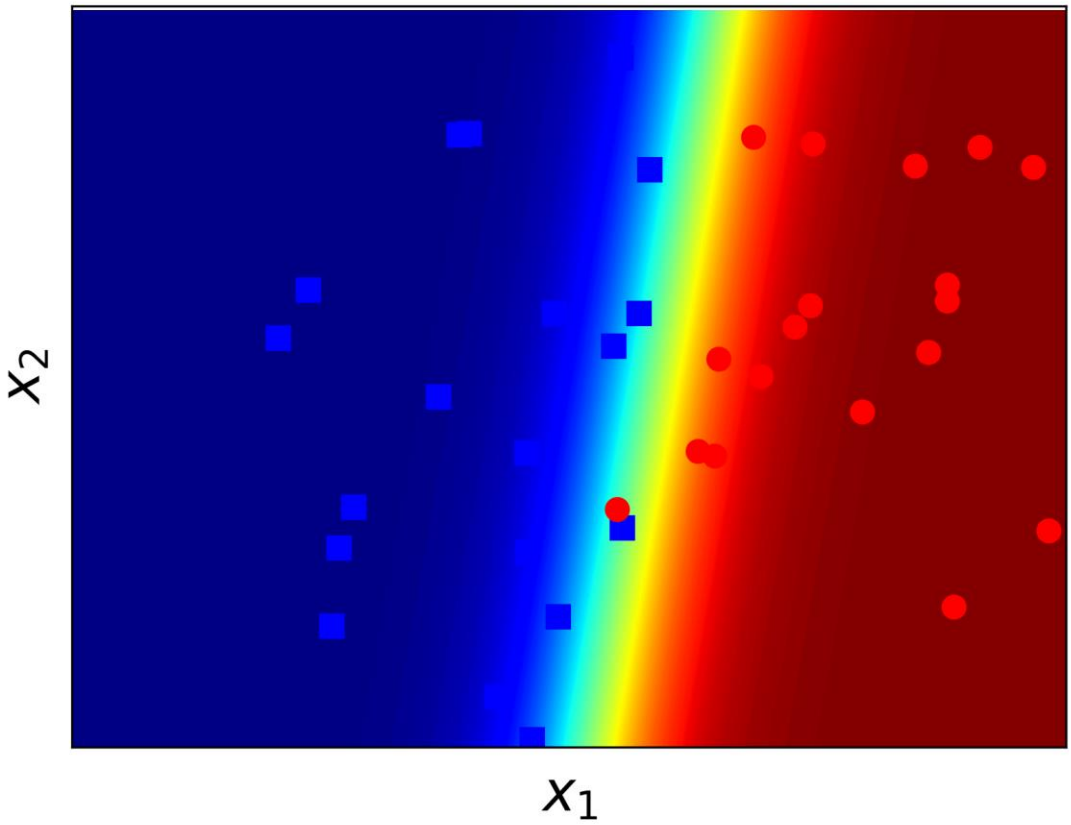
Bài 2:






```
[[ 1.        1.        1.        1.        1.        1.
   1.        1.        1.        1.        1.        1.
   1.        1.        1.        1.        1.        1.
   1.        1.        1.        1.        1.        1.
   1.        1.        1.        1.        1.        1.
   1.        1.        1.        1.        1.        1.
   1.        1.        1.        1.       ]
 [ 2.73951804  2.29913651  1.09236394  2.78601645  2.42092102  2.2706269
   1.33735121  0.92949127  1.96302398  2.87942062  1.85890055  2.93716989
   1.79640818  1.90878952  2.1077437   1.25914327  2.78049198  1.21777478
   2.44097816  2.27577431  3.71679844  3.49607296  4.54134755  4.36647759
```

```
  5.00776644  4.41283921  3.53549824  4.71709614  3.81621832  2.75974768
  3.30666588  4.44152932  5.08885396  5.75252276  4.57813385  4.08265249
  3.28484352  3.19495235  3.79964523  4.54084228]
 [ 2.16387243  0.03948643  2.46827551  1.18134403  2.34003074  1.58718284
  1.29522442  2.20558006  3.31157549  2.33999222 -0.66869682  3.11508225
  1.8912085   3.30199899  0.27066663  1.0758425   3.72338412  0.65047556
  0.70225602  1.04911367  2.26854343  3.29241247  2.49632113  3.13600779
  3.13099383 -0.2626343   2.00126215  3.23921368  3.25783314  1.28368162
  2.09541907  2.13038361  1.16768901  2.30071987  0.75695779  1.80897292
  1.570672    1.59635261  2.38464962  2.40779057]]
[[0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
 [0.]
```

[0.]

[0.]

[0.]

[0.]

[0.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]

[1.]]

[[-11.21325149]

[4.30832035]

[-0.75933172]]

C:\Users\ACER\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.10_qbz5n2kfra8p0\LocalCache\local-packages\Python310\site-packages\sklearn\linear_model_logistic.py:1173: FutureWarning: `penalty='none'`has been deprecated in 1.2 and will be removed in 1.4. To keep the past behaviour, set `penalty=None`.

  warnings.warn(

C:\Users\ACER\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.10_qbz5n2kfra8p0\LocalCache\local-packages\Python310\site-packages\sklearn\utils\validation.py:1141: DataConversionWarning: A column-vector y was passed when a 1d array was expected. Please change the shape of y to (n_samples,), for example using ravel().

  y = column_or_1d(y, warn=True)

[[-43.737504   17.34094286 -3.72826273]]

0.95

[[19 1]

 [1 19]]

Bài 3:

C:\Users\ACER\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.10_qbz5n2kfra8p0\LocalCache\local-packages\Python310\site-packages\sklearn\linear_model_logistic.py:458: ConvergenceWarning: lbfgs failed to converge (status=1):

STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:

https://scikit-learn.org/stable/modules/preprocessing.html

Please also refer to the documentation for alternative solver options:

https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression

  n_iter_i = _check_optimize_result(

[[0.37021425 0.04504631 0.12569399 0.42295425 0.46369397 0.55294594

1.7243153 0.85582429]]

Accuracy = 0.8714285714285714

Precission = 0.8741721854304636

Recall = 0.8354430379746836

b

MSE = 0.004641678440367083

c

Accuracy = 0.9

Precision = 0.9473684210526315

Recall = 0.8181818181818182

Bài 4:

[[3.65753143e-02 2.75763051e-03 -5.22466121e-01 1.53965741e-01

  1.47218340e-02 -9.26596183e-02 -9.84048218e-02 4.71615664e-02

 -7.33349727e-02 3.65753143e-02 1.44348605e+00 -2.21073712e+00

  3.57948285e+00 5.18882914e-01 -3.14893468e+00 -6.65638009e-02

 -2.20608744e-01 2.49659672e-01 -1.06073704e-02 1.49278634e-01

 -1.92821417e-01 -3.42975885e-02 3.80139291e-02 -4.02126788e-02

  7.16862405e-02 -2.68164308e-02 -5.82264604e-02 1.14472660e-01

  2.49131105e-04 8.56927476e-03 2.87553161e-02 -5.57437103e-02

 -1.61677199e-01 -5.29915489e-03 4.46124494e-05 3.85805101e-02

  1.82699270e-01 2.95765407e-01 -2.59190092e-01 1.55157026e-01

 -6.34004152e-01 5.15422440e-01]]

Accuracy_score: 0.9104151493080845

Recall_score: 0.6521739130434783

Precision_score: 0.3559322033898305

F1-score: 0.46052631578947373

Elapsed_time:0.5445380210876465[sec]

Accuracy_score: 0.832362223840738

Recall_score: 0.27289377289377287

Precision_score: 0.3367231638418079

F1-score: 0.3014668689934243

Elapsed_time:0.09899067878723145[sec]